Practice workbook · Measure food energy by burning and explain why the value is too low

Burning a food sample under a tube or can of water, describing the method, explaining heat losses and how a calorimeter or apparatus changes improve accuracy.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-FOOD-ENERGY-CALORIMETRY

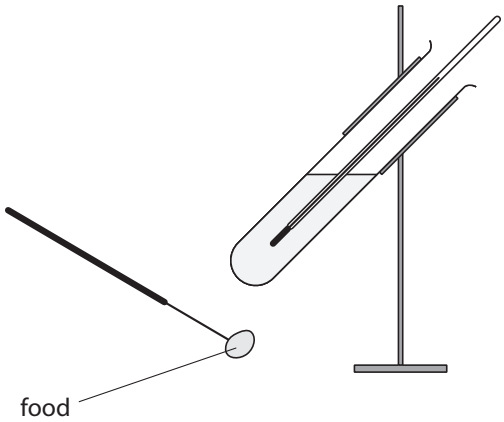
1. Question 7 · part a · 6 marks

November 2024 Paper 1

Worked-example retry · Attempt part a only.

7 In a healthy person, the amount of energy in the daily diet should equal the amount of energy used by the body.

(a) The equipment shown can be used to measure the amount of energy in food.



Describe how this equipment could be used to compare the energy content of different foods.

(6)

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1. Question 7 · part a · 6 marks

November 2024 Paper 1

Worked-example retry · Attempt part a only.

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(b) Describe a named method of transport that uses energy to move molecules across a cell membrane.

(2)

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(Total for Question 7 = 8 marks)

Working and answer

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2. Question 5 · part b.i · 4 marks

November 2020 Paper 1R

Further practice · Attempt part b.i only.

5 (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

(1)

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

(4)

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2. Question 5 · part b.i · 4 marks

November 2020 Paper 1R

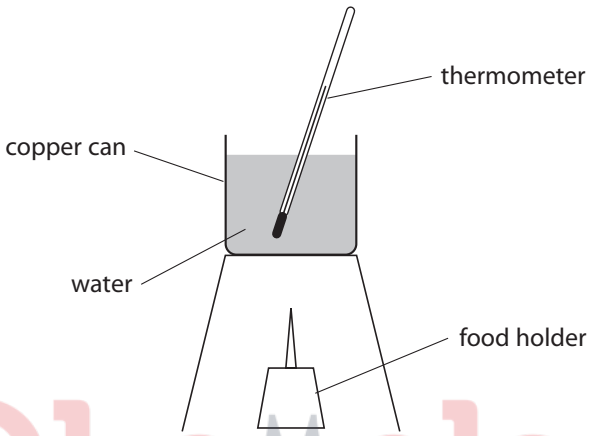
Further practice · Attempt part b.i only.

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(b) The diagram shows some apparatus that can be used to measure the amount of energy contained in different foods.



(i) Explain how the apparatus can be used to obtain data about the energy content of different foods.

(4)

(ii) Explain how this apparatus can be changed to improve the accuracy of the data collected for each food.

(3)

(Total for Question 5 = 12 marks)

2. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

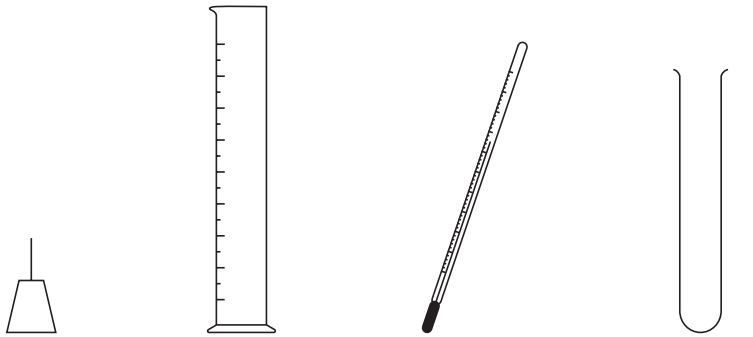
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3. Question 3 · part a.i · 5 marks

June 2022 Paper 1

Further practice · Attempt part a.i only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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3. Question 3 · part a.i · 5 marks

June 2022 Paper 1

Further practice · Attempt part a.i only.

(ii) Name two variables that need to be controlled in the investigation.

(2)

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(iii) Give two safety precautions needed for this investigation.

(2)

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3. Question 3 · part a.i · 5 marks · Working and answer

Working and answer

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4. Question 5 · part b.ii · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b.ii only.

5 (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

(1)

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

(4)

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4. Question 5 · part b.ii · 3 marks

November 2020 Paper 1R

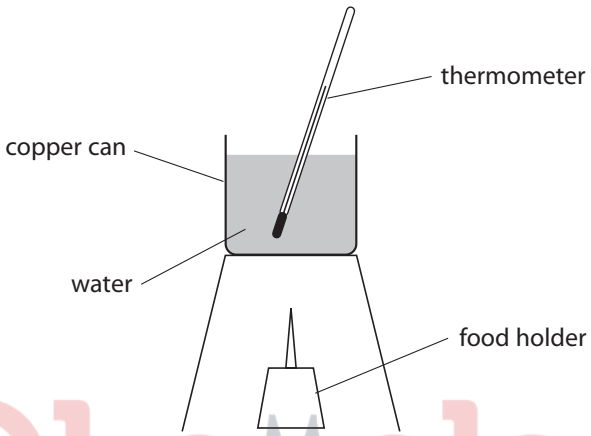
Further practice · Attempt part b.ii only.

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(b) The diagram shows some apparatus that can be used to measure the amount of energy contained in different foods.



(i) Explain how the apparatus can be used to obtain data about the energy content of different foods.

(4)

(ii) Explain how this apparatus can be changed to improve the accuracy of the data collected for each food.

(3)

(Total for Question 5 = 12 marks)

4. Question 5 · part b.ii · 3 marks · Working and answer

Working and answer

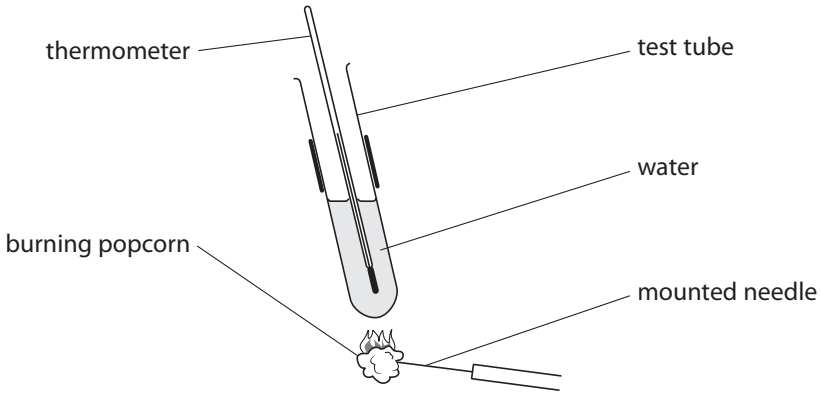
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5. Question 2 · part a.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part a.i only.

2 A student uses this apparatus to investigate the energy contained in popcorn.



This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

(1)

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(ii) Explain one safety precaution that the student should take during the investigation.

(2)

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5. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

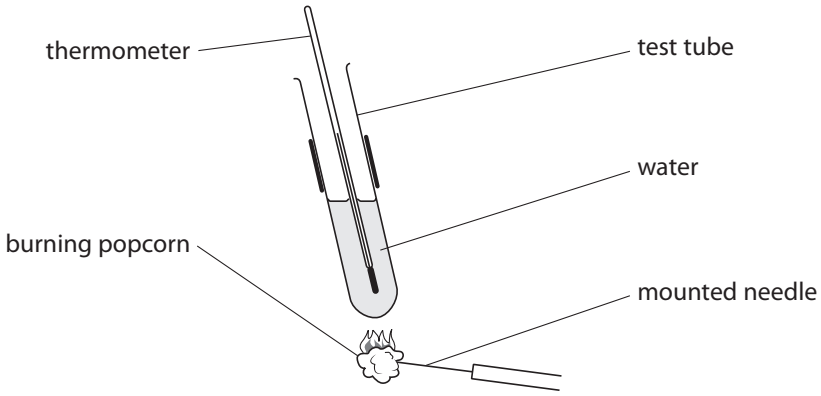
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6. Question 2 · part b.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.ii only.

2 A student uses this apparatus to investigate the energy contained in popcorn.



This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

(1)

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(ii) Explain one safety precaution that the student should take during the investigation.

(2)

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6. Question 2 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

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(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

(3)

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energy released = ... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

(3)

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6. Question 2 · part b.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.ii only.

(c) The student wants to compare the energy content of popcorn with the energy content of two other foods.

State four factors that the student should control to make sure the comparison is valid.

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(Total for Question 2 = 13 marks)

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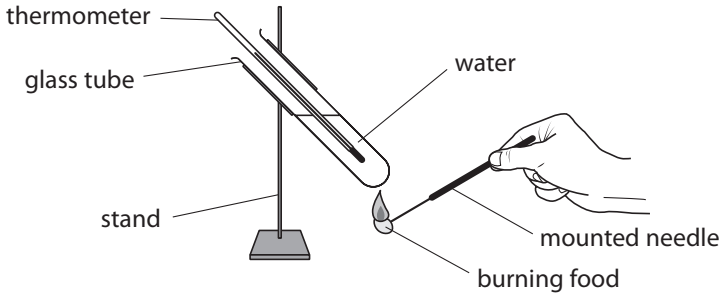
Working and answer

7. Question 7 · part a.iii · 5 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

7 A student uses this apparatus to investigate the energy content of different foods.



This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

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7. Question 7 · part a.iii · 5 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

(ii) State three factors that should be controlled during the investigation.

(3)

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

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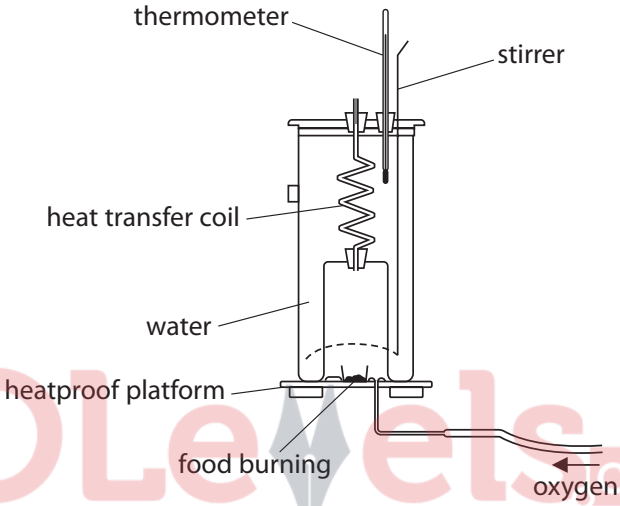
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7. Question 7 · part a.iii · 5 marks

June 2021 Paper 1

Further practice · Attempt part a.iii only.

(b) The diagram shows a calorimeter. This is a piece of apparatus that is used to measure the energy content of a sample of food accurately.



Discuss the features of this calorimeter that will give a more accurate measurement of the energy in a sample of food than the method used by the student.

(4)

(Total for Question 7 = 15 marks)

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7. Question 7 · part a.iii · 5 marks · Working and answer

Working and answer

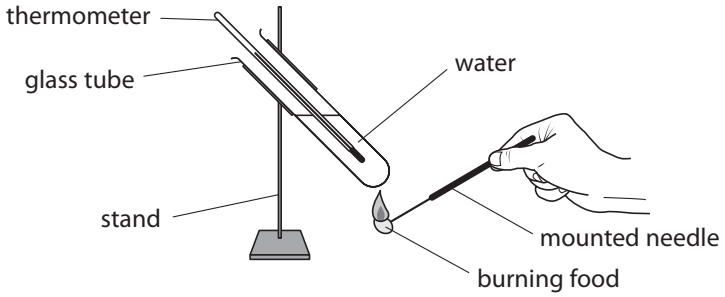
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8. Question 7 · part b · 4 marks

June 2021 Paper 1

Further practice · Attempt part b only.

7 A student uses this apparatus to investigate the energy content of different foods.



This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

(3)

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8. Question 7 · part b · 4 marks

June 2021 Paper 1

Further practice · Attempt part b only.

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(ii) State three factors that should be controlled during the investigation.

(3)

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

(5)

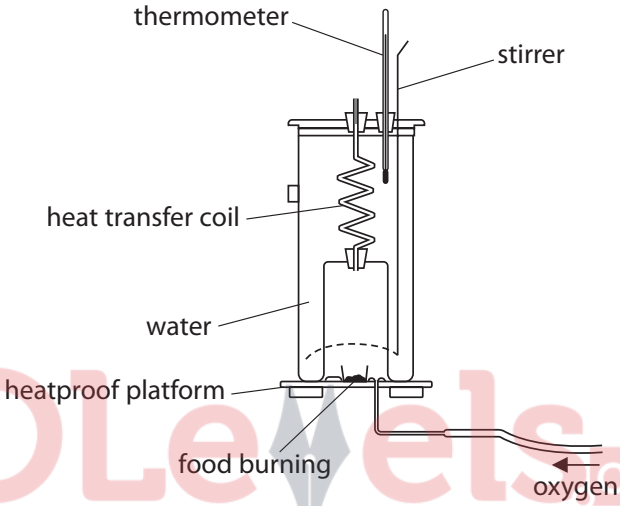
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8. Question 7 · part b · 4 marks

June 2021 Paper 1

Further practice · Attempt part b only.

(b) The diagram shows a calorimeter. This is a piece of apparatus that is used to measure the energy content of a sample of food accurately.



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(4)

(Total for Question 7 = 15 marks)

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8. Question 7 · part b · 4 marks · Working and answer

Working and answer

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